



# An Introduction to Naive Bayes Classifiers

How It Works, Where We See It, and Why It Matters

# Spams vs. Hams

congratulations!  
win a price...



hey friend! how  
are things...

I HAVE A DATE  
ON SUNDAY  
WITH WILL!!



Eh u remember  
how 2 spell his  
name...

WINNER!! As a valued  
network customer...

আমি গুগল-এর একজন  
নিয়োগ ব্যবস্থাপক।

Thanks for your  
subscription to...

Did you hear about  
the...

Wah lucky man...

Ok lar... Joking wif  
u oni...



Please call our  
customer service...

Tell where you  
reached...

Congrats! 1 year  
special...

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# Naive Bayes Models

- **Bayes' Theorem:** What we know to guess what's true.
- **Naive Assumption:** Assumes every word is independent
- **Use cases:** *text classification, sentiment analysis, spam filtering.*

# Naive Bayes Models

A *probability-based* classifier  
that assumes all features work  
*independently*.

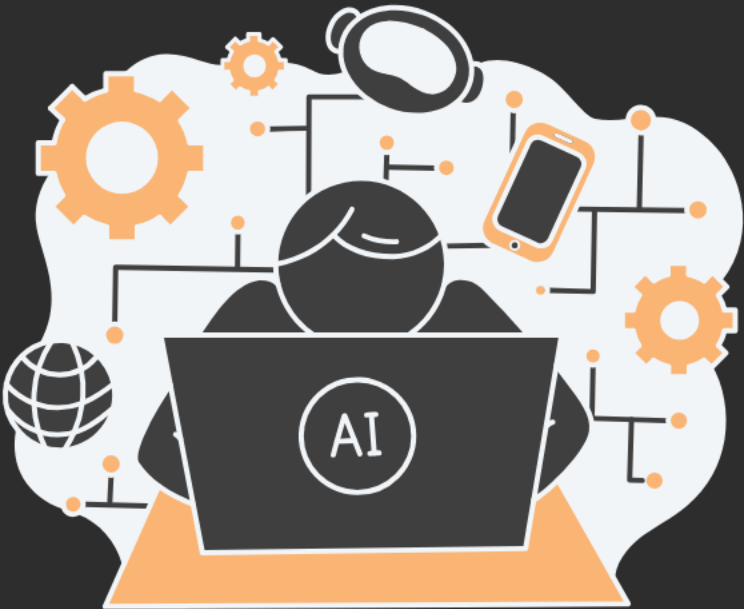


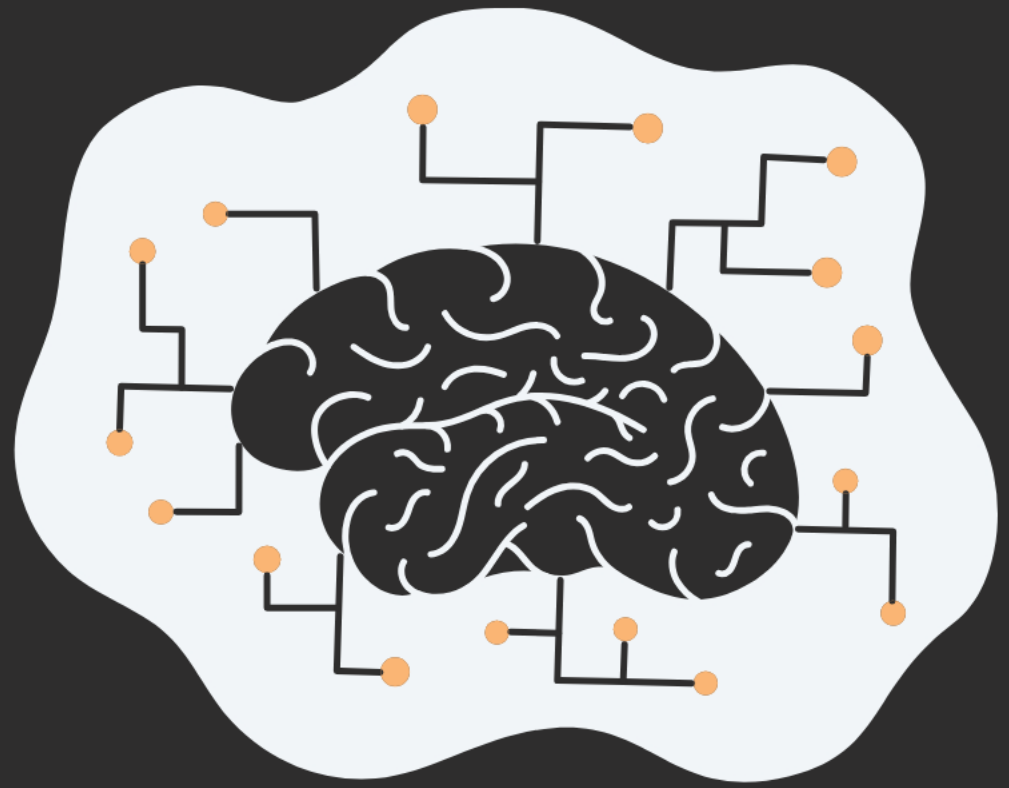
# Three Naive Bayes Models

Multinomial Naive Bayes

Bernoulli Naive Bayes

Gaussian Naive Bayes





# Multinomial Naive Bayes



## Count Matters

Perfect for text data  
where we care about  
**how many times a  
word appears**



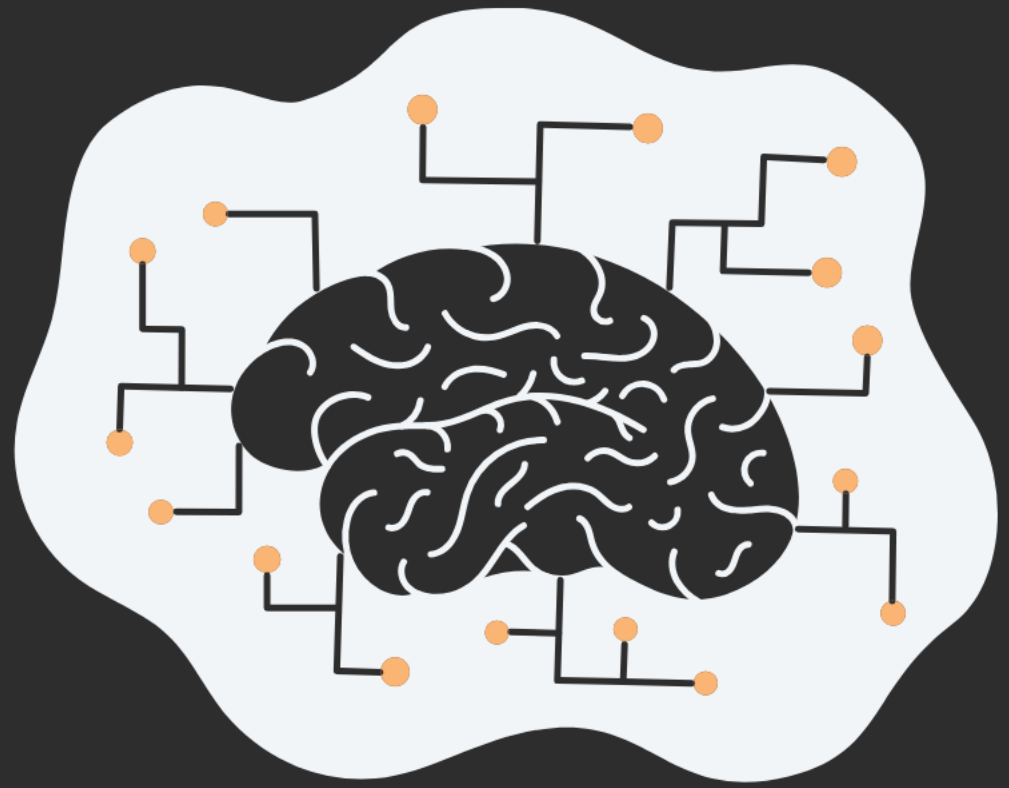
## Example

The word *free* shows  
up 5 times → **5**



# Practical Example

| Message                   | Words                  | Count                               |
|---------------------------|------------------------|-------------------------------------|
| Free prize now            | "free", "prize", "now" | free: 1<br>prize: 1<br>now: 1       |
| Free free unlimited offer | "free", "offer"        | free: 2<br>unlimited: 1<br>offer: 1 |
| Hello friend, hello       | "hello", "friend"      | hello: 2<br>friend: 1               |

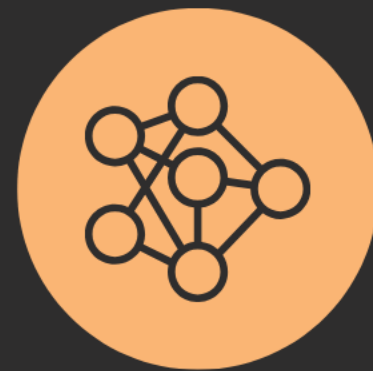


# Bernoulli Naive Bayes



## Presence Matters

Doesn't care how many times a word appears; **just whether it's there or not**

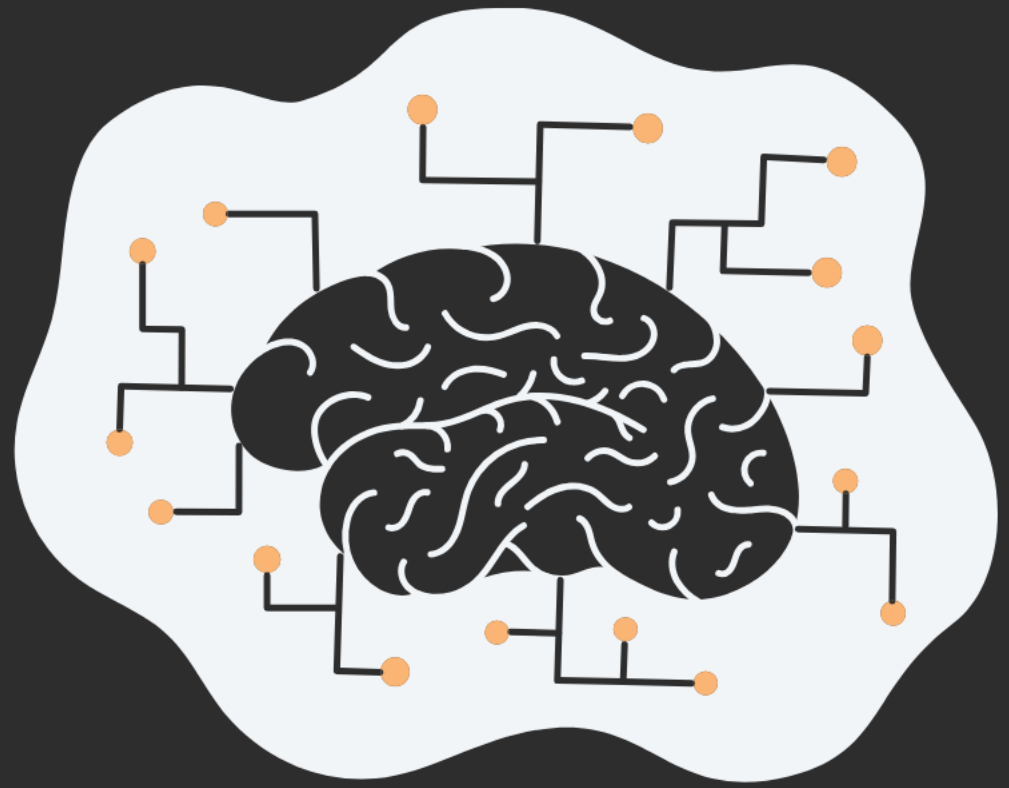


## Example

The word *free* shows up 5 times → **1** (Present)

# Practical Example

| Message                   | Words                  | Presence                            |
|---------------------------|------------------------|-------------------------------------|
| Free prize now            | “free”, “prize”, “now” | free: 1<br>prize: 1<br>now: 1       |
| Free free unlimited offer | “free”, “offer”        | free: 1<br>unlimited: 1<br>offer: 1 |
| Hello friend, hello       | “hello”, “friend”      | hello: 1<br>friend: 1               |



# Gaussian Naive Bayes



## Normal Distribution

Numerical data that follows a **normal distribution**



## Use Cases

*Continuous numbers,  
sensor readings,  
pixel intensities*

# Practical Example

| Message                   | Words                  | TF-IDF Scores                                |
|---------------------------|------------------------|----------------------------------------------|
| Free prize now            | “free”, “prize”, “now” | free: 0.82<br>prize: 0.54<br>now: 0.36       |
| Free free unlimited offer | “free”, “offer”        | free: 0.91<br>unlimited: 0.62<br>offer: 0.48 |
| Hello friend, hello       | “hello”, “friend”      | hello: 0.73<br>friend: 0.41                  |

# Final Recap

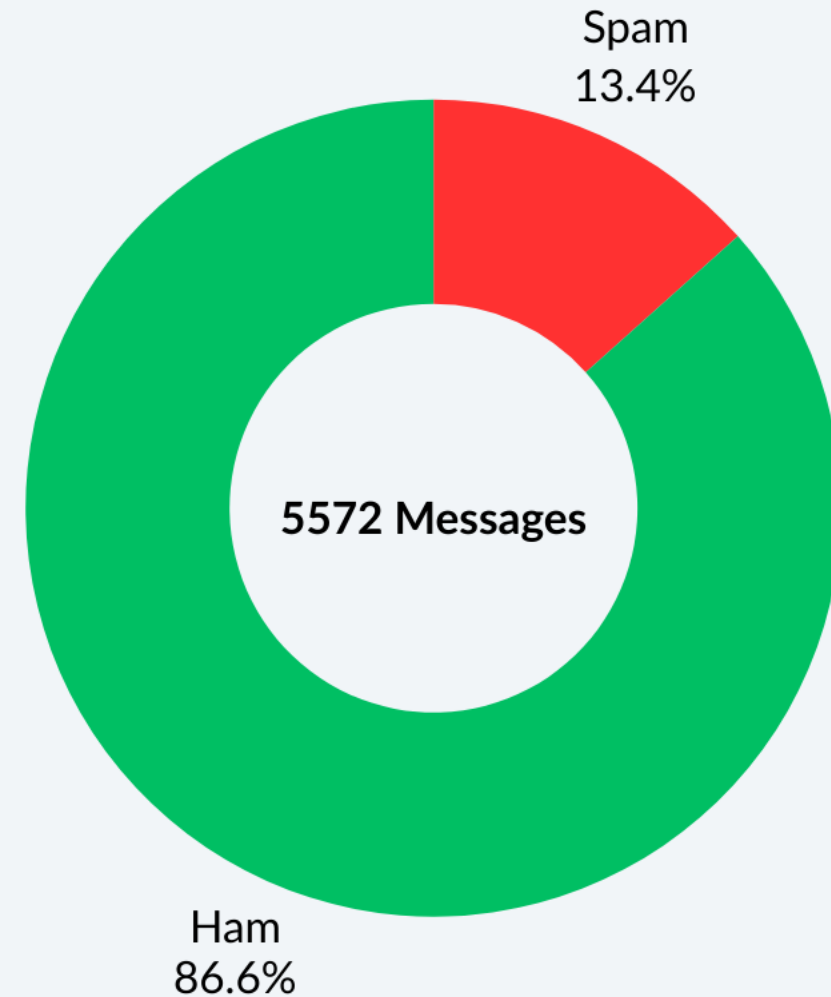
| Model                | Input Type        | What it Looks for                    |
|----------------------|-------------------|--------------------------------------|
| <b>MultinomialNB</b> | Word Counts       | Frequency of each word               |
| <b>BernoulliNB</b>   | Binary Values     | Whether a word appears or not        |
| <b>GaussianNB</b>    | Continuous Values | Statistical distribution of features |



# SMS Spam Dataset

## Sample Dataset

- **Ham:** U dun say so early hor... U c already then say...
- **Ham:** Ok lar... Joking wif u oni...
- **Spam:** Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005.
- **Spam:** WINNER!! As a valued network customer you have been selected to receivea £900 prize



# Report for MultinomialNB

**Accuracy Score: 98.68%**

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.99      | 1.00   | 0.99     | 1448    |
| 1            | 0.97      | 0.93   | 0.95     | 224     |
|              |           |        |          |         |
| accuracy     |           |        | 0.99     | 1672    |
| macro avg    | 0.98      | 0.96   | 0.97     | 1672    |
| weighted avg | 0.99      | 0.99   | 0.90     | 1672    |

# Report for BernoulliNB

**Accuracy Score: 97.19%**

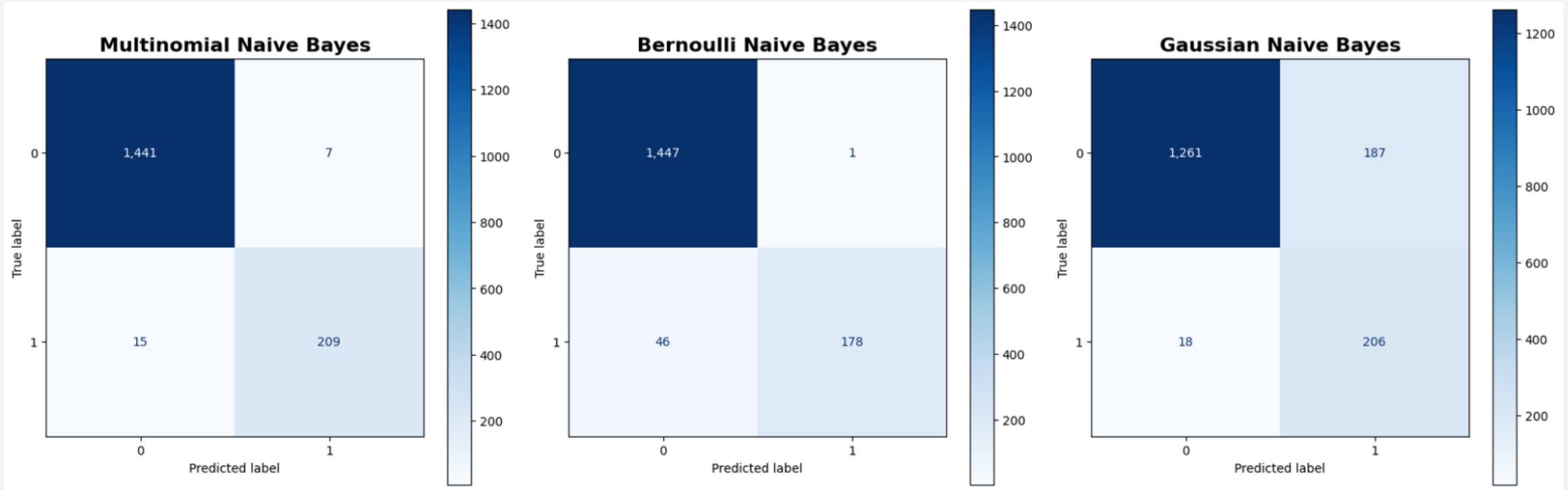
|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.97      | 1.00   | 0.98     | 1448    |
| 1            | 0.99      | 0.79   | 0.88     | 224     |
|              |           |        |          |         |
| accuracy     |           |        | 0.97     | 1672    |
| macro avg    | 0.98      | 0.9    | 0.93     | 1672    |
| weighted avg | 0.97      | 0.97   | 0.97     | 1672    |

# Report for GaussianNB

**Accuracy Score: 87.74%**

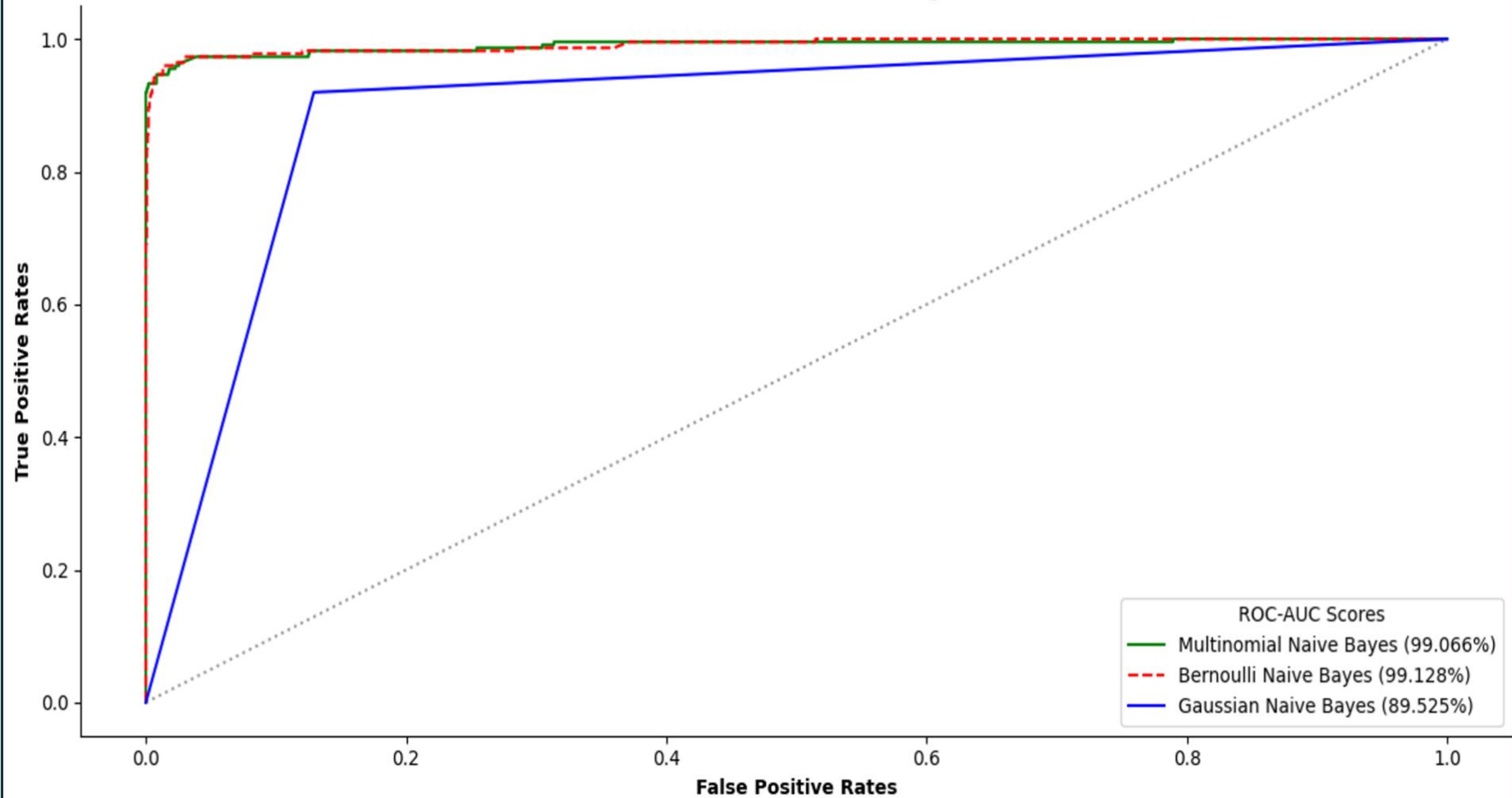
|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.99      | 9.87   | 0.92     | 1448    |
| 1            | 0.52      | 0.92   | 0.67     | 224     |
|              |           |        |          |         |
| accuracy     |           |        | 0.88     | 1672    |
| macro avg    | 0.76      | 0.90   | 0.8      | 1672    |
| weighted avg | 0.52      | 0.98   | 0.89     | 1672    |

# Confusion Matrices

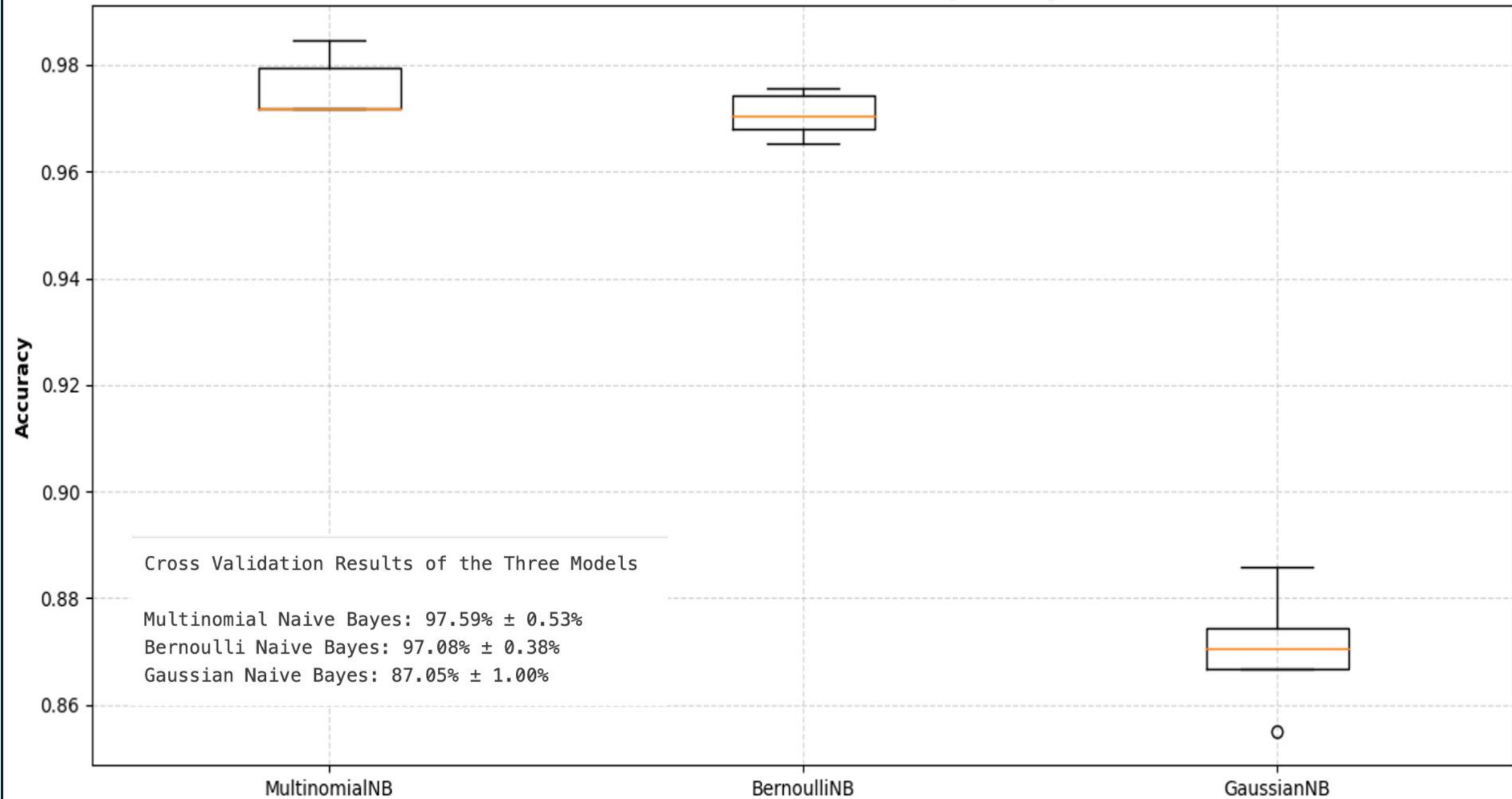


- **Minimize False Positives** → Bernoulli Naive Bayes
- **Minimize False Negatives** → Multinomial Naive Bayes
- **Overall Best Model** → Multinomial Naive Bayes
- **Overall Worst Model** → Gaussian Naive Bayes

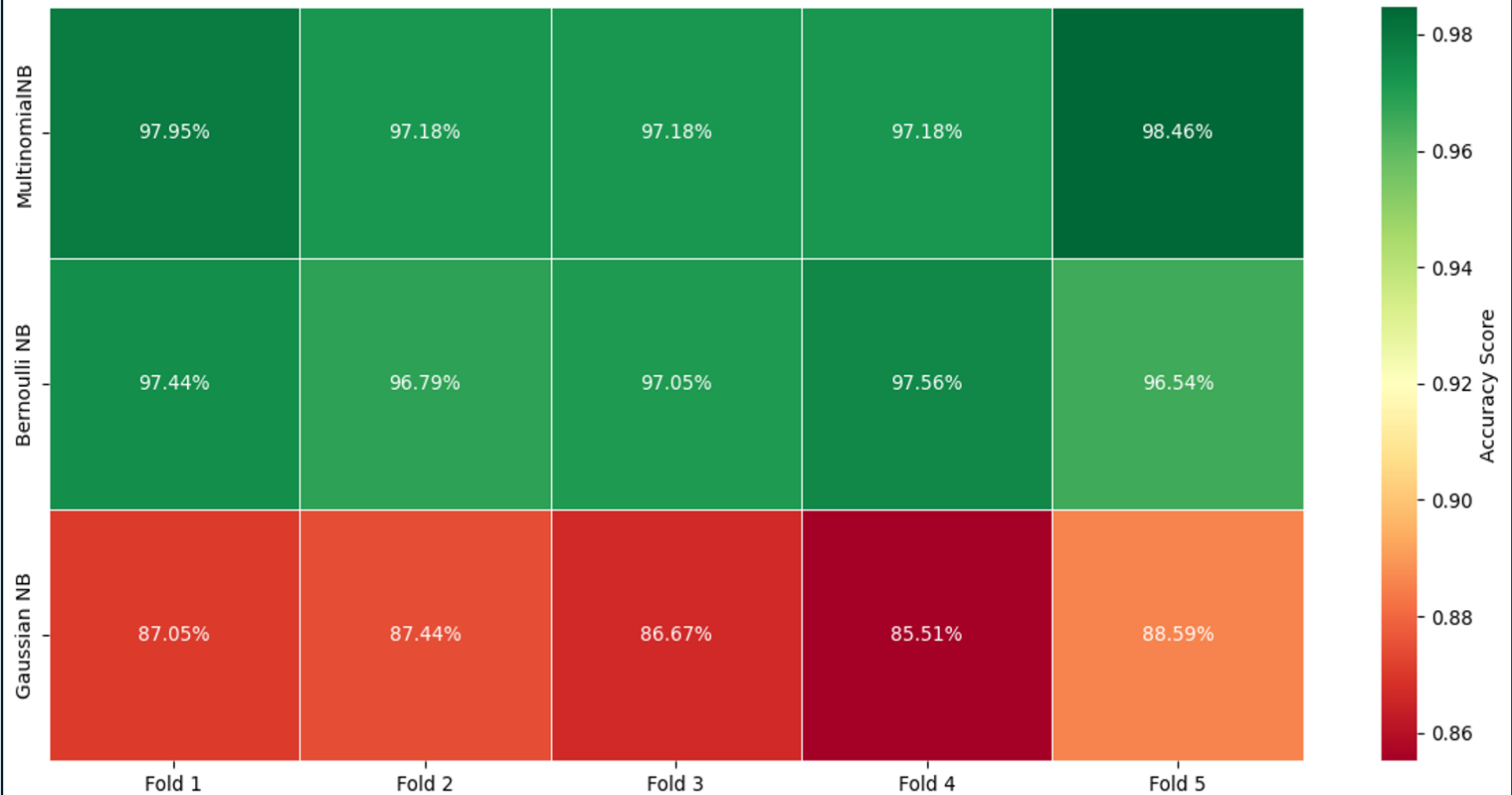
# ROC Curves of the Naive Bayes Models



# 5-Fold Cross-Validation Accuracy Comparison

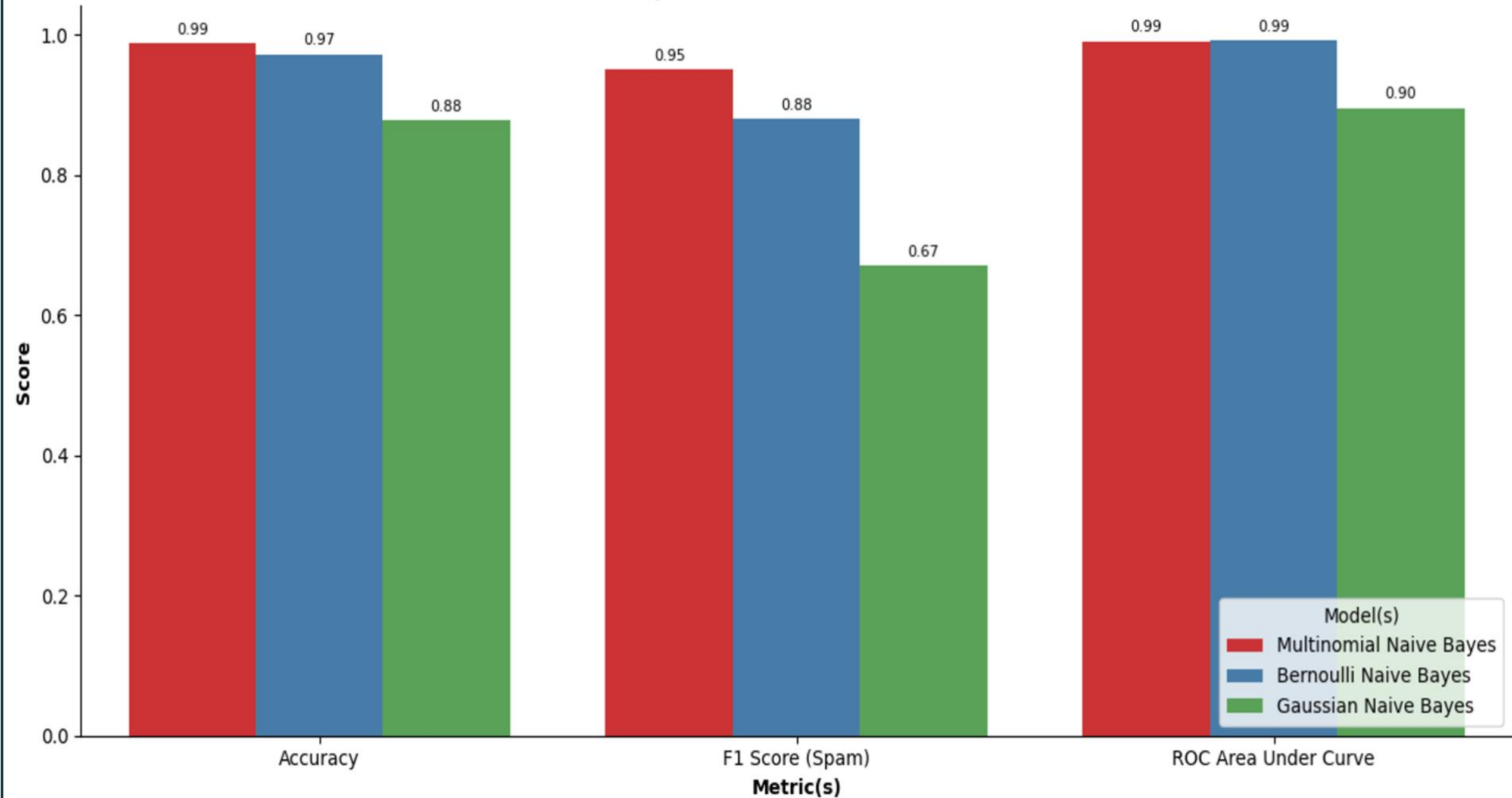


## Cross-Validation Folds





# Model Comparison across Three Metrics



# Best Model: Multinomial NB

**Accuracy**

98.68%

**Precision**

97.00%

**Recall**

93.00%

**F1-Score**

95.00%

**ROC-AUC**

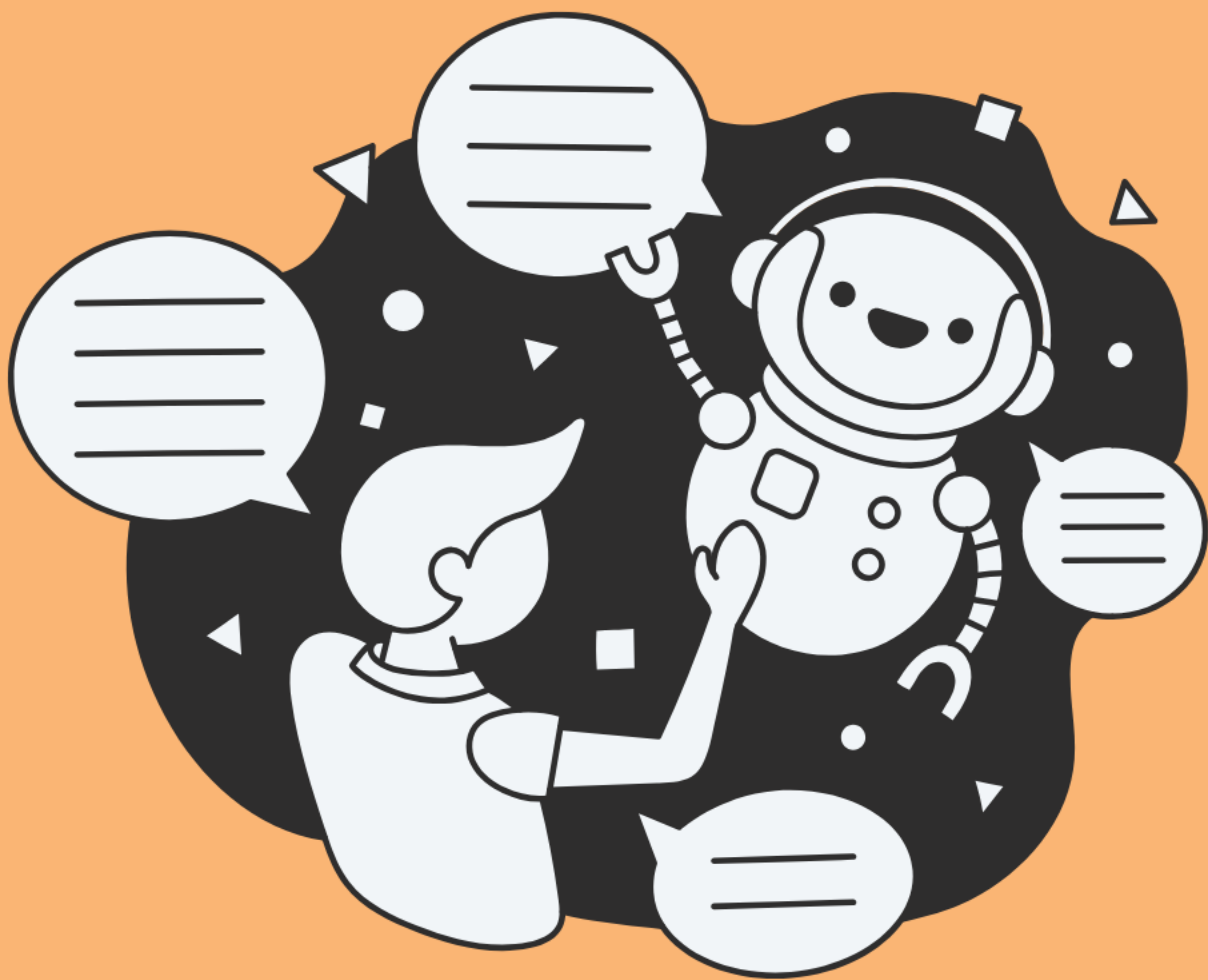
99.07%

**FPR Rate**

0.48%

**FNR Rate**

6.70%



# The End

Thank you for  
listening!